

Directed numbers on the number plane



1

a (6, 7)

b (3, 9)

c (4, 4)

d (9, 5)

e (-4, 3)

f (-9, 6)

g (-2, 7)

h (-7, 8)

i (-2, -8)

j (-6, -7)

k (-7, -2)

l (-5, -9)

m (3, -2)

n (4, -7)

o (8, -6)

p (7, -1)

2

a (1.5, 3)

b (-1.5, 0)

c (-2.5, -3.5)

d (3.5, -3.5)

3

a
i 0

ii 0

b
i 0

ii 0

4 y -axis

5 $a = 0$

6

a Q

b S

c P

d R

7

a $A(3, 0)$

b $B(-2.5, 0)$

c x -axis

8

a 1st quadrant

b 2nd quadrant

c 4th quadrant

d 3rd quadrant

9

a 3rd quadrant, 4th quadrant

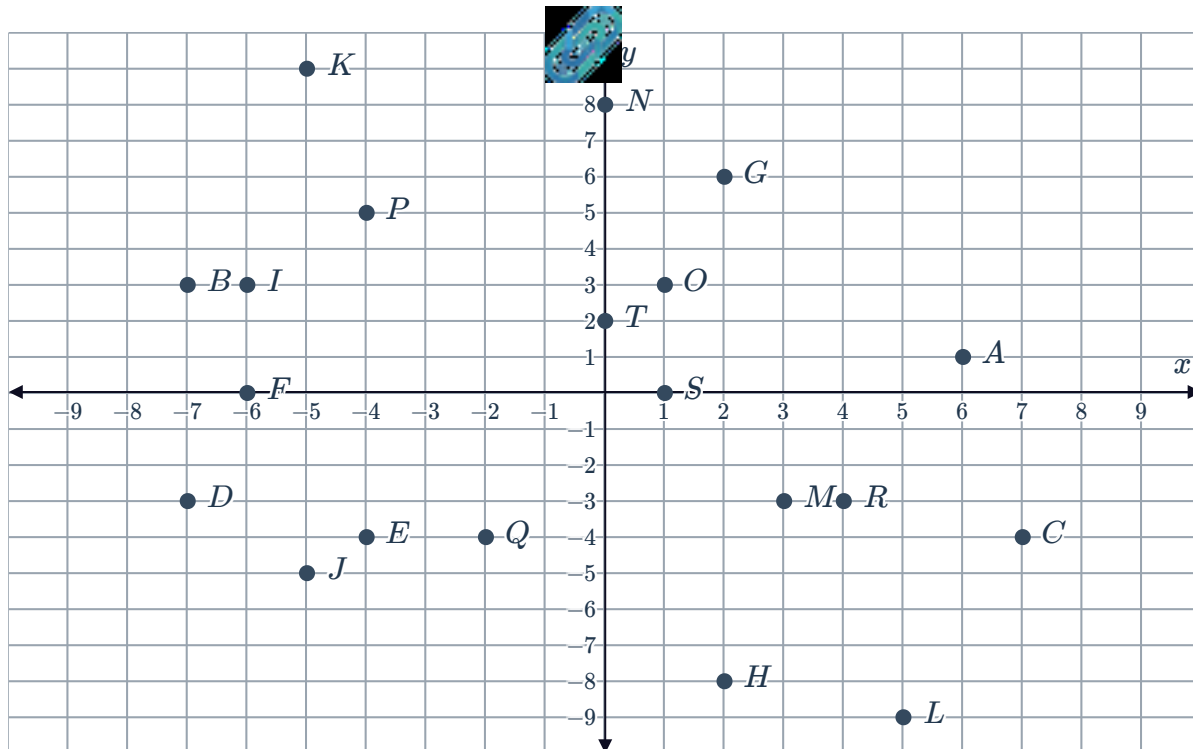
b 2nd quadrant, 3rd quadrant

c 1st quadrant, 3rd quadrant

d 2nd quadrant

e 3rd quadrant

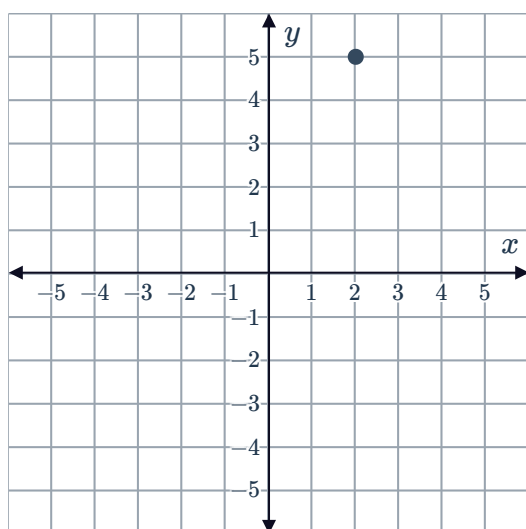
10



11

a

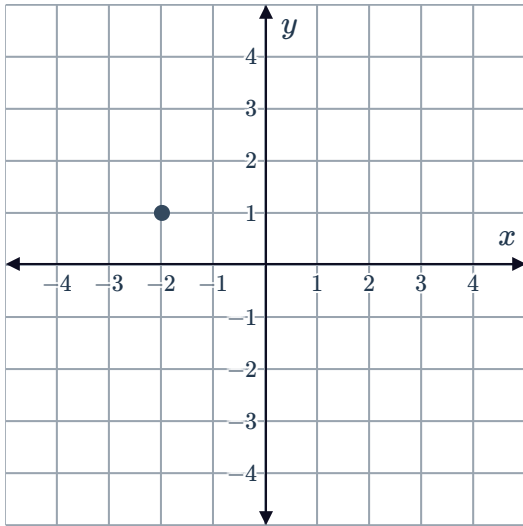
i



ii 1st quadrant

b

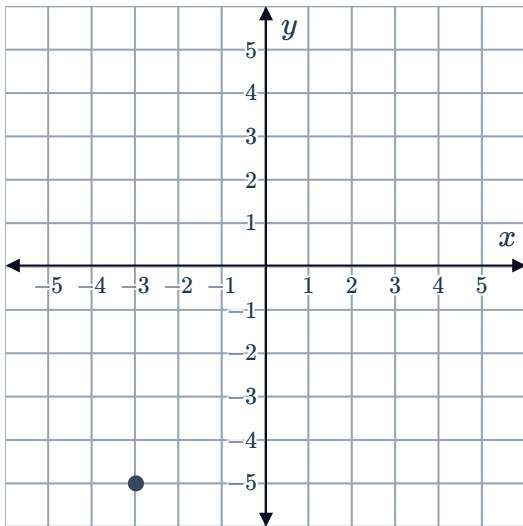
i



ii 2nd quadrant

c

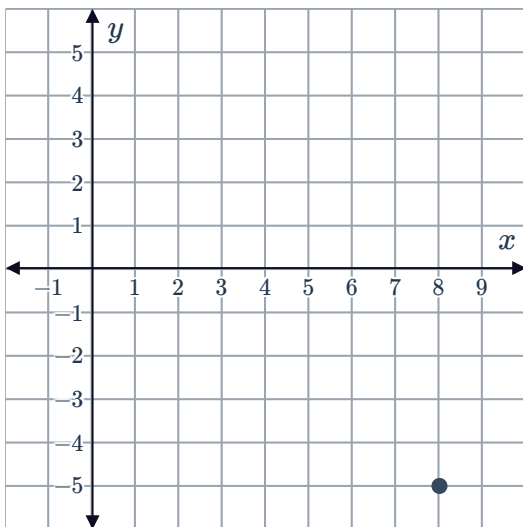
i



ii 3rd quadrant

d

i

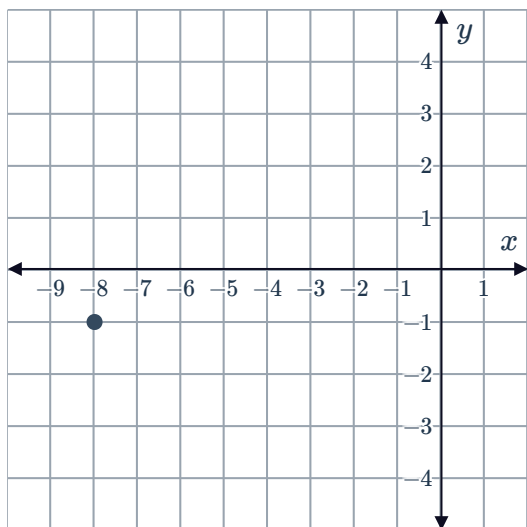


ii 4th quadrant

e
i

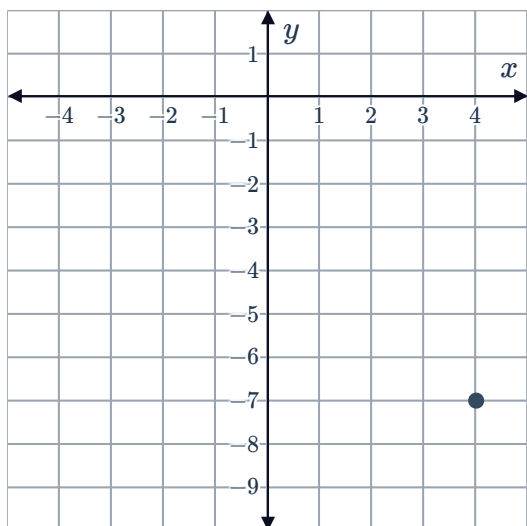


ii 3rd quadrant



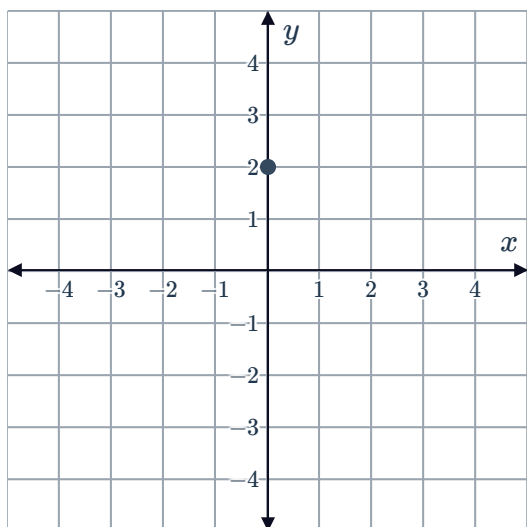
f
i

ii 4th quadrant



g
i

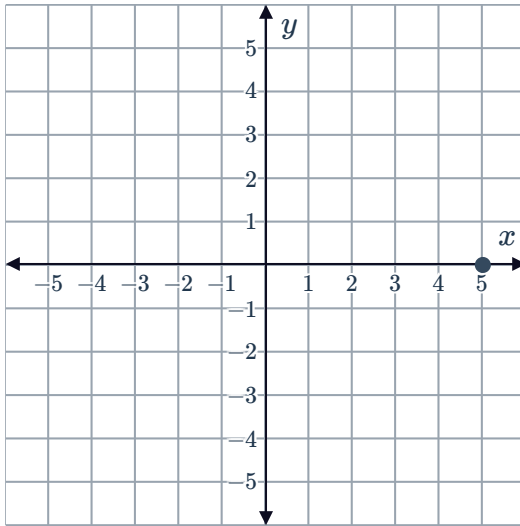
ii No quadrant



h
i

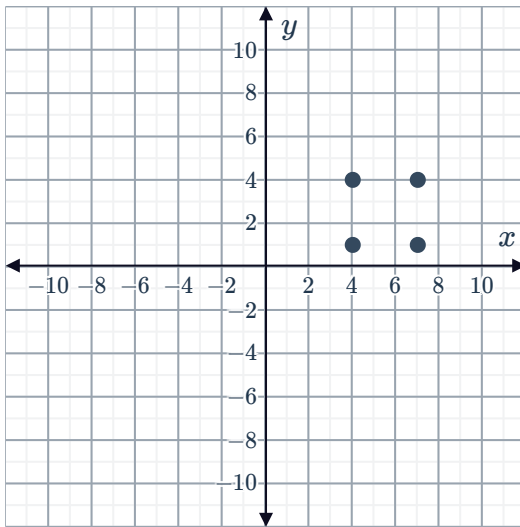


ii No quadrant



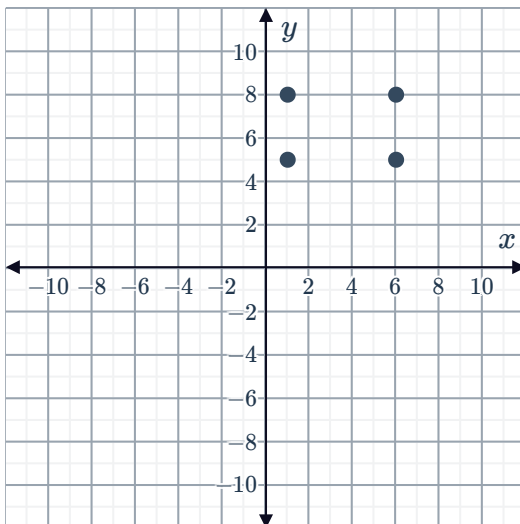
12
a
i

ii Square

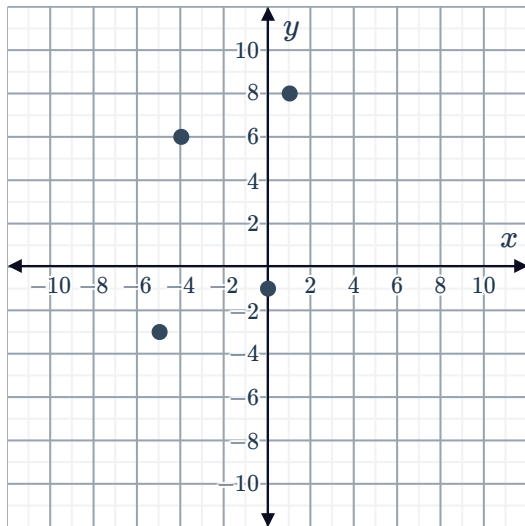


b
i

ii Rectangle

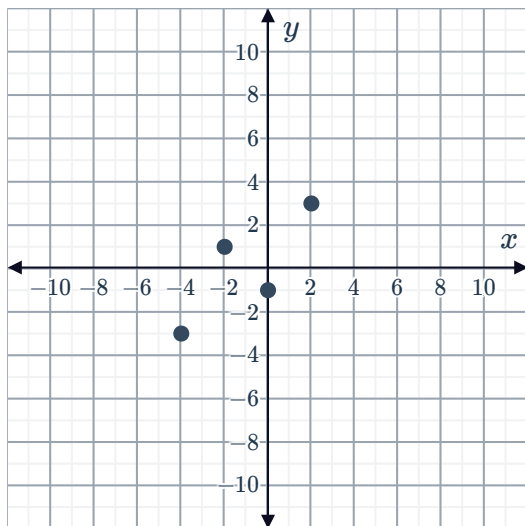


c i



ii Parallelogram

d i



ii Rhombus

13

a $A(5, 9), B(5, -3), C(-2, -3), D(-2, 9)$

b $T(8, 6), U(8, -9), V(-4, -9)$

c $A(6, 2), B(-9, 3), C(-9, -7), D(6, -8)$

d $S(8, 9), T(-5, 4), U(-7, -3), V(6, -6)$

14

a Move 2 units to the right, and then 4 units down.

b Move 9 units to the right, and then 14 units up.

15

a $(0, -9)$

b $(-3.5, 0)$

c $(-7, 6)$

d $(6, -2)$

e $(4, 3)$

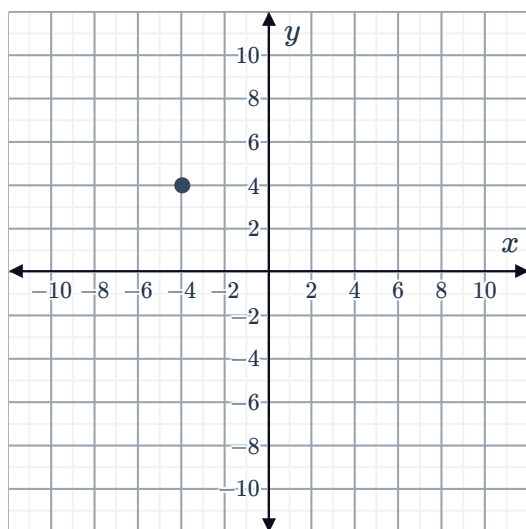
f $(-2, 1)$

16 B

17 1



18
a



b $(-4, 4)$

Distance between two points

19

a 3 units

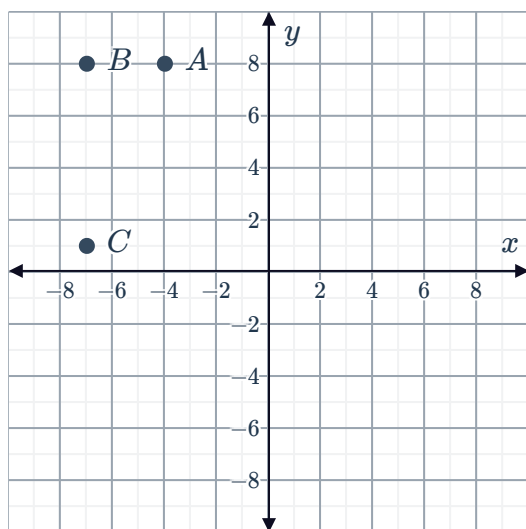
b 8 units

c 4 units

d 9 units

20

a

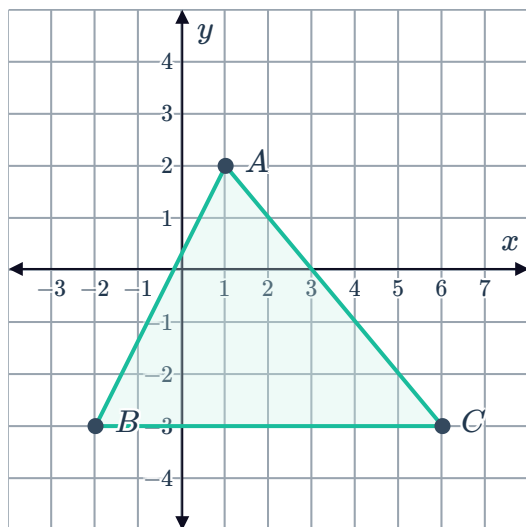


b 3 units

c 7 units

21

a



b 5 units

c 8 units

d 20 units^2

Geometric applications

22 $(-2, 2)$

23 $(-2, -1)$